



INTERACTIVE VISUALIZERS

Tool: VisuAlgo

CO3-AT3

Insert(80)

80 has been inserted!

GUIDED BY:
DR. M. Shakila

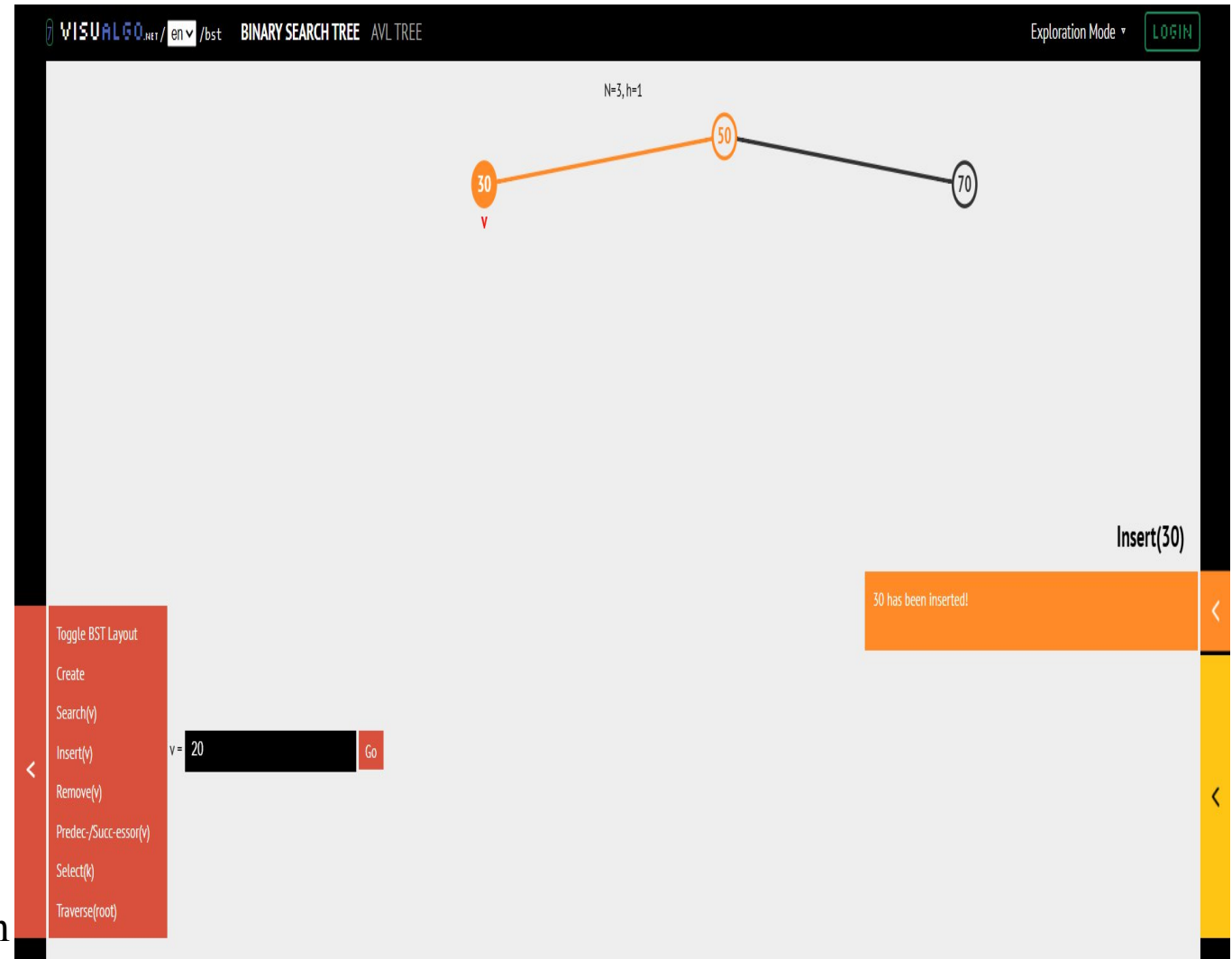
PRESENTED BY:
Bhavya S(192524376)

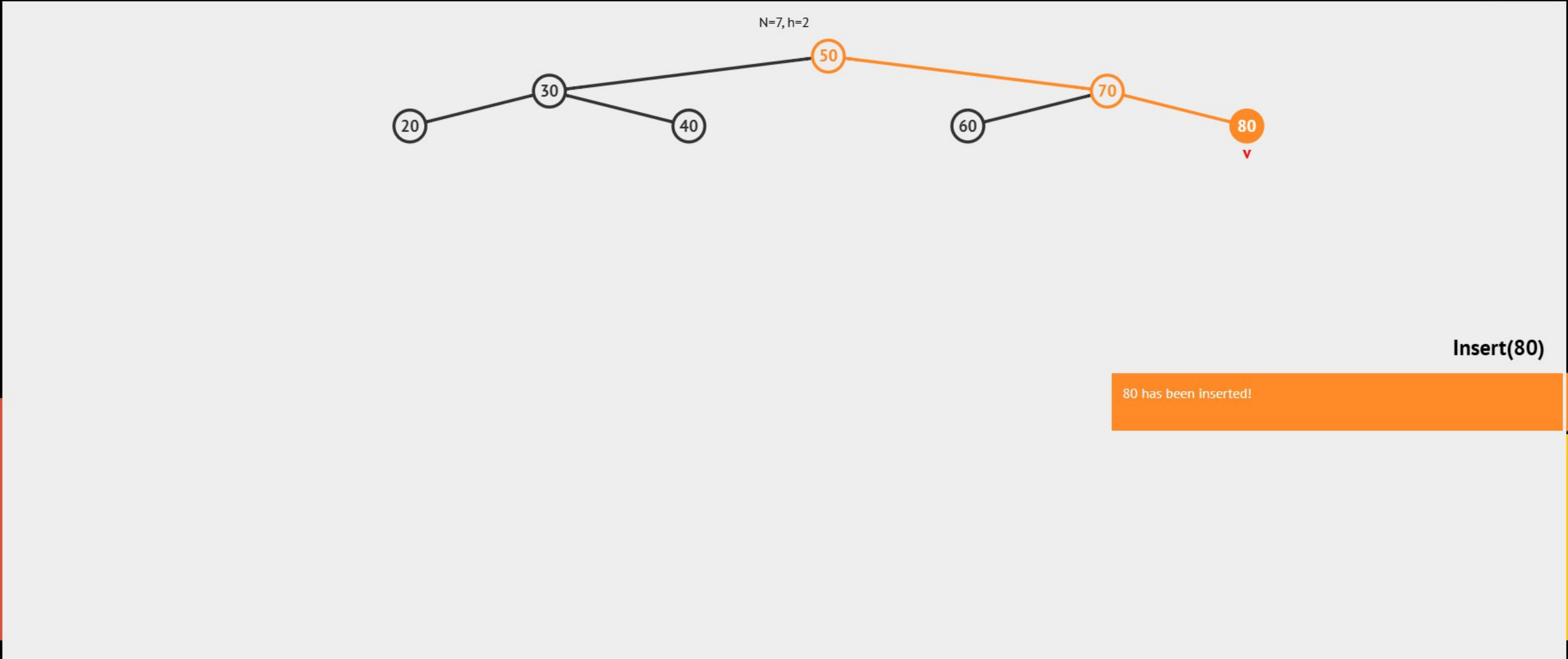
OPERATIONS:

1. INSERTION
2. DELETION
3. SEARCHING
4. TRAVERSING

1.INSERTION:

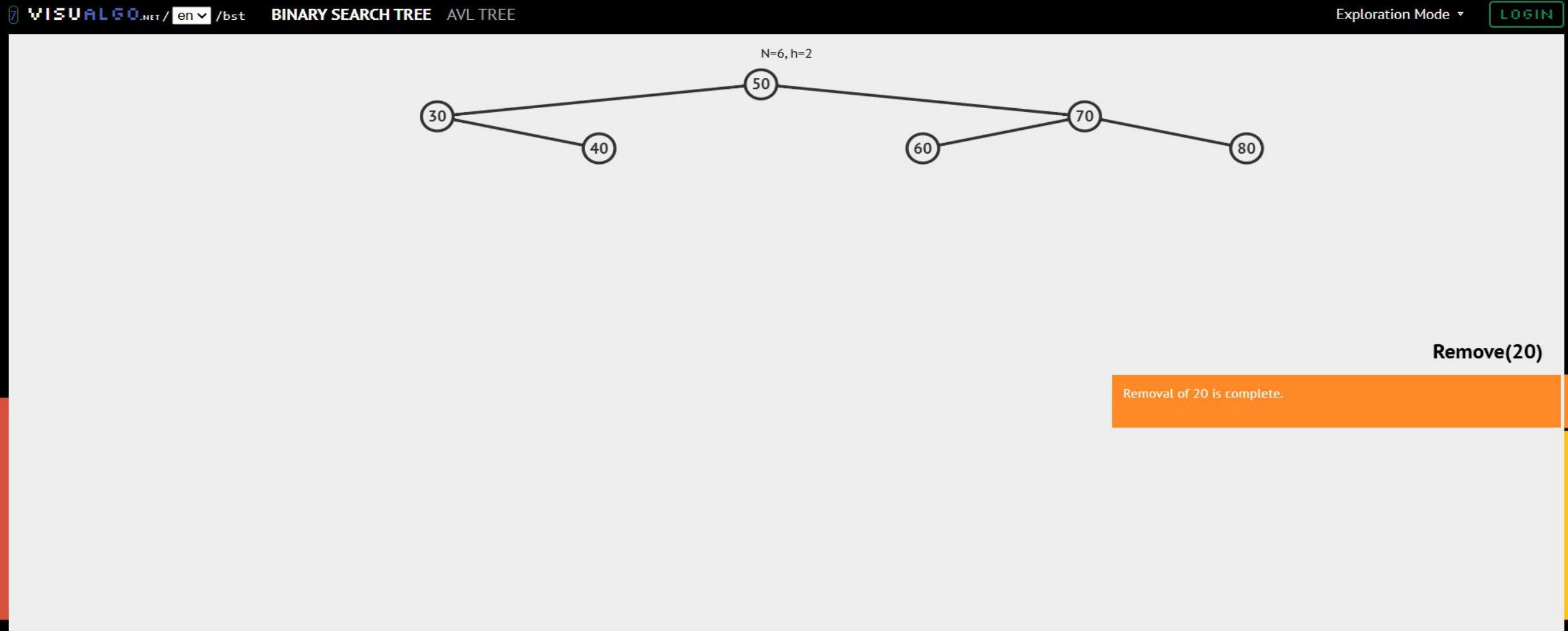
- Open the VisuAlgo BST simulator and select Insert.
- Enter 50 and click Insert to create the root node.
- Enter 30, 70, 20, 40, 60, 80 one by one and click Insert after each value.
- Observe the simulator automatically placing each value according to the BST rule.
- The final simulated tree is 50, 30, 70, 20, 40, 60, 80 with the correct BST structure.





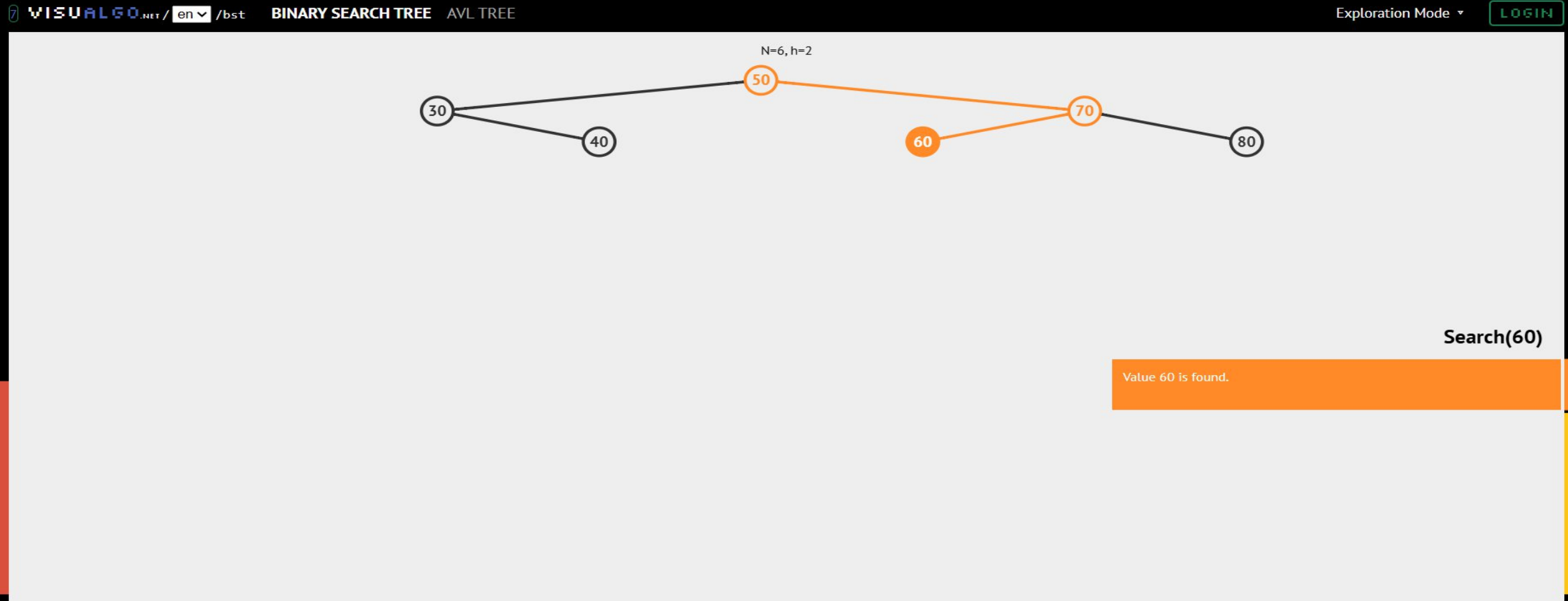
2.DELETION:

- Select Delete in the BST simulator.
- Enter 20 and click Delete.
- The simulator searches for 20 using the path: $50 \rightarrow 30 \rightarrow 20$.
- Node 20 is a leaf node because it has no children, so it is directly removed.



3.SEARCHING:

- Select Search in the BST simulator.
- Enter the value 60 and click Search.
- The simulator first compares 60 with 50; since $60 > 50$, it moves to the right.
- It then compares 60 with 70; since $60 < 70$, it moves to the left and reaches 60.
- 60 is found successfully. The search path is $50 \rightarrow 70 \rightarrow 60$.



4.TRAVERSING:

i. Inorder: (Left - Root - Right)

Visualgo.net interface showing a Binary Search Tree (BST) with 7 nodes and height 2. The tree structure is as follows:

```
graph TD; 50((50 root)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

The interface includes a sidebar with the following menu items:

- Toggle BST Layout
- Create
- Search(v)
- Insert(v)
- Remove(v)
- Predec-/Succ-essor(v)
- Select(k)
- Traverse(root)

The bottom toolbar shows the following traversal options:

- Inorder(root) (selected)
- Preorder(root)
- Postorder(root)

The code block on the right displays the Inorder traversal logic:

```
if this is null
    return
Inorder(left)
visit this
Inorder(right)
```

Navigation controls at the bottom include a zoom slider (1.0x to 1x) and a progress bar.

ii. Preorder: (Root - Left - Right)

Visualgo.net interface showing a Binary Search Tree (BST) with 7 nodes and height 2. The tree structure is as follows:

```
graph TD; 50((50 root)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

The interface includes a sidebar with the following menu items:

- Toggle BST Layout
- Create
- Search(v)
- Insert(v)
- Remove(v)
- Predec-/Succ-essor(v)
- Select(k)
- Traverse(root)

Below the menu, there are buttons for traversal types: Inorder(root), Preorder(root), and Postorder(root). The Preorder(root) button is currently selected.

Inorder Traversal

Inorder traversal of the whole BST is complete.
Visitation sequence: 20,30,40,50,60,70,80.

```
if this is null
  return
Inorder(left)
visit this
Inorder(right)
```

At the bottom, there is a playback control bar with a 1.0x speed indicator and navigation buttons.

iii. Postorder: (Left - Right - Head)

Binary Search Tree, AVL Tree - V

Ask Gemini

visualgo.net/en/bst?utm_source=chatgpt.com

All Bookmarks

VISUALGO.NET / en / bst

BINARY SEARCH TREE AVL TREE

Exploration Mode

LOGIN

N=7, h=2

```
graph TD; 50((50 root)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

<

Toggle BST Layout

Create

Search(v)

Insert(v)

Remove(v)

Predec-/Succ-essor(v)

Select

Traverse(root)

Preorder Traversal

Preorder traversal of the whole BST is complete.
Visitation sequence: 50,30,20,40,70,60,80.

```
if this is null
    return
visit this
Preorder(left)
Preorder(right)
```

>

Thank
you!